

TABLE I
Latest Instruments in the Last One Year

Instrument	Model	Manufacturer	Features
All-in-one	VirtualBench	National Instruments	- Contains a mixed-signal oscilloscope with protocol analysis, digital multimeter, function generator, programmable DC power supply and digital I/O - Connects display via existing Wi-Fi network or directly to a network hosted by VirtualBench - Small form factor of 254×190.5×73.5mm
All-in-one	MD03000	Tektronix	- Contains a mixed-domain oscilloscope, spectrum analyser, logic analyser, protocol analyser, digital voltmeter and an arbitrary function generator - Completely customisable; enables customers to select the functionality and performance, and is fully upgradable
All-in-one	WaveSurfer 3000	Teledyne LeCroy	- Contains an AWG, a protocol analyser and digital voltmeter - CAN and LIN trigger and decode capabilities help analyse and debug automotive systems using CAN and/or LIN serial data communication standards - Enables import of .csv files to recreate analogue waveforms
Arbitrary waveform generator	AG4151	OWON	- Maximum 150MHz frequency output - Up to 400MSa/s sample rate and 32-bits frequency resolution - Vertical resolution: 14-bits, up to 1M arbitrary waveform length - Sine, square, ramp, pulse, noise, exponential rise and decay, 32-channel digital waveform, DC and user-defined arbitrary waveform
Arbitrary waveform generator	AG1022F	OWON	- Maximum 25MHz frequency output 125MSa/s sample rate, and 32-bits frequency resolution - Vertical resolution: 14-bits, 8K arbitrary waveform length - Sine, square, ramp, pulse, noise, exponential rise, exponential fall, $\sin(x)/x$, step wave and others, total 26 built-in waveforms and other user-defined arbitrary waveforms
Arbitrary waveform generator	AFG1022	Tektronix	- Dual-channel, 25MHz and up to 10 Vp-p output amplitude - Four run modes, 50 built-in frequently-used waveforms and built-in 200MHz frequency counter - Targets education field and entry-level testing
Vector signal analyser with vector signal generator (optional)	MS2830A	Anritsu	- Supports multi-function vector signal analyses in both time and frequency domains - Power consumption of 110VA minimum - Also supports an analogue signal generator, audio analyser and modulation analysis
Vector signal generator	VSG25A	Signal Hound	- Weights 130gm and fits into a pocket - Frequency range of 100MHz to 2.5GHz, output amplitude from -40dBm to +10dBm and 100MHz of modulation bandwidth - Supports modulation for BPSK, QPSK, DQPSK, OQPSK, $\pi/4$ DQPSK, 8PSK, 16PSK, 16QAM, 64QAM and 256QAM protocols
Vector signal generator	TSG4100A	Tektronix	- VSG at an affordable RF signal generator price - Easily upgraded in the field for more advanced vector- and digital-modulation capabilities - Targets product design, testing and manufacturing
Vector signal transceiver	NI PXIe-564X	National Instruments	- Vector signal analyser (VSA) and generator in a single module 65MHz to 6GHz frequency range, up to 200MHz instantaneous bandwidth 24 channels of high-speed digital I/O up to 250Mbit/s

shape. The bandwidth of AWGs is usually limited compared to function generators due to the complex techniques used to generate these signals.

Sophisticated signal generators are also divided into the following categories based on their application:

Audio frequency (AF) signal generators. These are optimised for use in AF range of 20Hz to 20kHz

and above, and are used for measuring distortion and checking frequency response in audio equipment.

Radio frequency (RF) signal generators. These are used to generate RF signals and are generally used in design and test applications in the RF range, particularly telecommunications.

Vector signal generators (VSGs).

These are basically RF signal generators with advanced modulation formats for complex waveforms.

Pulse generators. These produce output in the form of pulses of logic 1s and logic 0s with different analogue characteristics such as rise time, fall time and delays.

When these create mere pulses without controllable parameters, these are known as logic generators. These are used to generate pulses that stimulate logic circuits.

Technology trends

In April 2015, an article published on *Electronics B2B* (electronicsb2b.com), titled 'Higher efficiency rates and customisation options driving signal generator market,' mentions the trends in the signal generator industry as:

Customisation.

The instrument can be customised based on application requirements.

Compactness. Developers are trying to shrink the size of the device without compromising on the effectiveness of measurements.

USB interfaces. These make the instrument compatible with various devices to be tested and enable seamless transfer of data.

Support for communication technologies. The gadget has to support